message would be signed, timestamped, and delivered through a uniform channel, providing the same assurances of provenance and integrity that SSF offers across organizational boundaries.

This internal use of the SSF model further streamlined ACME's enterprise PIP. It changed how context is shared by using an event bus instead of custom API calls, offering a consistent way to deliver reliable, verifiable signals wherever they are needed.

12.6.4 The broader lesson

ACME's experience shows that normalization acts as a link between policy creation and dependable enforcement. Policies can't be consistent if their inputs aren't, and they can't scale if each team has to build its own context system. By centralizing the work of collecting, verifying, and structuring context, the identity team transformed authorization data into a shared enterprise service, one that minimized engineering overhead and sped up the adoption of fine-grained authorization across the organization.

Normalization doesn't make policies smarter; it makes them more trustworthy. A clear schema and dependable signal processing give decision engines the ability to evaluate policies accurately, no matter where or how they're deployed.

12.7 Putting it all together

Dynamic authorization depends on context, and context depends on signals that are accurate, secure, and well-structured. Over the course of this chapter, we've seen how ACME learned that collecting signals is only half the challenge—making them consistent, verifiable, and usable is what turns authorization into a dependable service.

Each dimension of the three-dimensional choice space—the signals themselves, how they're delivered, and how fresh they are—affects the assurance and performance of every authorization decision. Signals define what the system knows; delivery determines how that information travels; and freshness defines how long it can be trusted.